Controller Remote Connection Guide - OpenVPN
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macOS Software requirements:

- OpenVPN (`brew install openvpn`)
    - To start openvpn now and restart at startup: `sudo brew services start openvpn`
    - The location of the `openvpn` binary needs to be added to the shell `PATH`. Depending on the shell in use (typically `bash` or `zsh`), you will need to create or modify `.zshrc` or `.bashrc` at in the user folder, i.e `~/.zshrc`
    - Within the config file add the following line
        - `export PATH=$PATH:/usr/local/opt/openvpn/sbin`
        - Save and close the file. The `PATH` will take effect when the shell is restarted or:
        - Enable the new `PATH` immediately with command `source .zshrc`

OpenVPN Configuration::

- Files needed (Access to these files is restricted as they contain secure keys. Ask Christopher Coté first, Sean Murphy second for access)
    - `connect-local.bin`
    - `connect.bin`
    - `ovpn-fwds-client.crt`
    - `ovpn-fwds-client.key`
    - `ovpn.conf`
    - `ovpn.crt`
    - `station (.pem)`

- Open `ovpn.conf` with a text editor and enter the correct paths for `ca, cert` and `key` match the location on your system. See example below

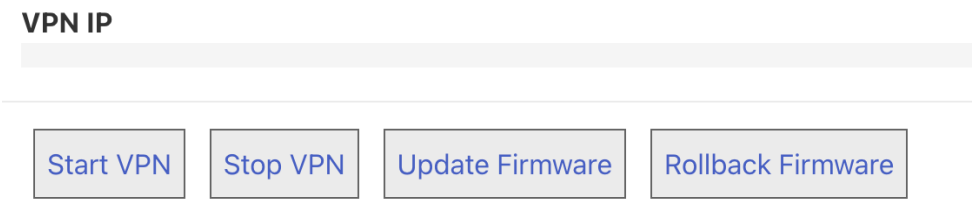
```
ovpn.conf
;http-proxy [proxy server] [proxy port #]

# Wireless networks often produce a lot
# of duplicate packets.  Set this flag
# to silence duplicate packet warnings.
;mute-replay-warnings

# SSL/TLS parms.
# See the server config file for more
# description.  It's best to use
# a separate .crt/.key file pair
# for each client.  A single ca
# file can be used for all clients.


ca /Users/smurph/Documents/Frontline/Remote/ovpn.crt
cert /Users/smurph/Documents/Frontline/Remote/ovpn-fwds-client.crt
key /Users/smurph/Documents/Frontline/Remote/ovpn-fwds-client.key


# Verify server certificate by checking that the
# certificate has the correct key usage set.
# This is an important precaution to protect against
# a potential attack discussed here:
#  http://openvpn.net/howto.html#mitm
#
```

- Copy `station` file to `~/.ssh` directory. Create directory if needed
- Change permission of `Station` file with `chmod 600 ~/.ssh/station`
- Edit `ovpn.conf` file to add paths to `.crt` & `.key` files for `ca, cert` and `key` entries
- Copy `connect-local.bin` & `connect.bin` to `/usr/local/bin` (or add location of files to PATH)
    - Note: `connect-local.bin` is to connect controllers on a local network. The default LAN IP range is: `192.168.0.`**`$1`** If your LAN has a different IP range, you will need to edit the `connect-local.bin` script
- Make `connect-local.bin` & `connect.bin` executable with `chmod 755` *`filename`*
- Run `sudo openvpn --config /`**`your/path/to`**`/ovpn.conf` to start openvpn
- Successful connection will look like this
- When finished the openvpn session can be terminated with `CTRL-C`.

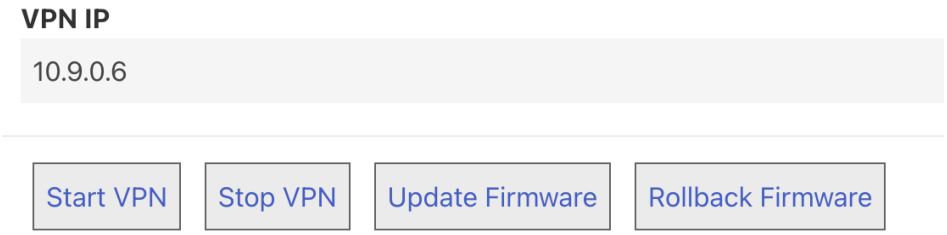
Connection to remote Controller:

- Open the Frontline Controller API Dashboard
- Find the desired Controller ID and click through the Controller Dashboard

- At the bottom of the dashboard find the following buttons:



- Click "Start VPN" / Wait up to 1 min for the dashboard to refresh
- On VPN connection you should see:



- Note the last digit of the IP address: `10.9.0.`**`x`**
- Open a new terminal window (leave the session running `openvpn` alone)
- Enter `connect.bin` **`x`** Where 'x' is the digit(s) to the right of the last group in the IP address. In the example above the command is `connect.bin 6`
- Successful connection should show a login screen as expected via a console connection to the Controller. See example here
- When you have finished with the remote controller, close the session with the command `exit`
- If finished with VPNs, terminate the openvpn session from it's terminal window with `CTRL-C`

Sample output of successful connection to openvpn:

```
2023-07-12 14:05:47 WARNING: Compression for receiving enabled. Compression has been
used in the past to break encryption. Sent packets are not compressed unless
"allow-compression yes" is also set.
2023-07-12 14:05:47 DEPRECATED OPTION: --cipher set to 'BF-CBC' but missing in
--data-ciphers (AES-256-GCM:AES-128-GCM:CHACHA20-POLY1305). OpenVPN ignores --cipher
for cipher negotiations.
2023-07-12 14:05:47 WARNING: file
'/Users/smurph/Documents/Frontline/Remote/ovpn-fwds-client.key' is group or others
accessible
2023-07-12 14:05:47 OpenVPN 2.6.5 x86_64-apple-darwin22.4.0 [SSL (OpenSSL)] [LZO]
[LZ4] [PKCS11] [MH/RECVDA] [AEAD]
2023-07-12 14:05:47 library versions: OpenSSL 3.1.1 30 May 2023, LZO 2.10
2023-07-12 14:05:47 WARNING: No server certificate verification method has been
enabled.  See http://openvpn.net/howto.html#mitm for more info.
2023-07-12 14:05:47 TCP/UDP: Preserving recently used remote address:
[AF_INET]52.207.97.101:1196
2023-07-12 14:05:47 Socket Buffers: R=[786896->786896] S=[9216->9216]
2023-07-12 14:05:47 UDPv4 link local: (not bound)
2023-07-12 14:05:47 UDPv4 link remote: [AF_INET]52.207.97.101:1196
```

```
2023-07-12 14:05:47 TLS: Initial packet from [AF_INET]52.207.97.101:1196, sid=b0de9d14
601a51ae
2023-07-12 14:05:47 VERIFY OK: depth=1, CN=FWDS CA
2023-07-12 14:05:47 VERIFY OK: depth=0, CN=fwds-server
2023-07-12 14:05:47 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384,
peer certificate: 2048 bit RSA, signature: RSA-SHA256
2023-07-12 14:05:47 [fwds-server] Peer Connection Initiated with
[AF_INET]52.207.97.101:1196
2023-07-12 14:05:47 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2023-07-12 14:05:47 TLS: tls_multi_process: initial untrusted session promoted to
trusted
2023-07-12 14:05:47 PUSH: Received control message: 'PUSH_REPLY,route 10.9.0.0
255.255.255.0,topology net30,ping 10,ping-restart 120,ifconfig 10.9.0.10
10.9.0.9,peer-id 1,cipher AES-256-GCM'
2023-07-12 14:05:47 OPTIONS IMPORT: --ifconfig/up options modified
2023-07-12 14:05:47 OPTIONS IMPORT: route options modified
2023-07-12 14:05:47 Opened utun device utun7
2023-07-12 14:05:47 /sbin/ifconfig utun7 delete
ifconfig: ioctl (SIOCDIFADDR): Can't assign requested address
2023-07-12 14:05:47 NOTE: Tried to delete pre-existing tun/tap instance -- No Problem
if failure
2023-07-12 14:05:47 /sbin/ifconfig utun7 10.9.0.10 10.9.0.9 mtu 1500 netmask
255.255.255.255 up
2023-07-12 14:05:47 /sbin/route add -net 10.9.0.0 10.9.0.9 255.255.255.0
add net 10.9.0.0: gateway 10.9.0.9
2023-07-12 14:05:47 Initialization Sequence Completed
2023-07-12 14:05:47 Data Channel: cipher 'AES-256-GCM', peer-id: 1, compression: 'lzo'
2023-07-12 14:05:47 Timers: ping 10, ping-restart 120
```

Sample output of successful connection to remove Controller:

```
Last login: Wed May  5 18:28:38 2021 from 10.9.0.1
#######################################################################
#
#   Frontline Wildfire Defense System Controller
#   ------------------------------------
#   Configuration Commands:
#       setup-station
#         - first command for a new controller
#         - also run to change the primary interface
#       setup-system
#         - second command for a new controller
#         - also run to:
#           o change the zone configuration
#           o enable/disable cellular interface
#           o enable/disable satellite interface
#
#       IMPORTANT NOTE: always use the 'reboot' command after running
#       either or both of these commands.
#
#   Diagnostic Commands:
#       cellular-check
#         - tests the cellular modem operation
#         - prints test status and signal level to the console
```

```
#       satellite-check -m
#         - runs the monitor mode satellite test
#         - after 10 minutes prints the satellite visibility uptime
#         - use the -f option to print full statistics
#       satellite-check -t
#         - runs the loopback mode satellite test
#         - run monitor mode test first to verify satellites visibility
#       satellite-check -r
#         - runs the receive mode satellite test
#         - requires user to initiate server request when prompted
#
#       IMPORTANT NOTES:
#         - Diagnostics require proper system configuration. Please
#           ensure that both the setup-station and the setup-system have
#           been run and that the system has been rebooted.
#         - The controller should be operating on the primary network
#           interafce when running any diagnostics. Failure to have the
#           primary network interface properly configured will interfere
#           with diagnostic operation.
#
#   Informational Commands:
#       sat-imei
#         - prints the satellite modem IMEI to the console
#       cell-imei
#         - prints the cellular modem IMEI to the console
#       cell-ccid
#         - prints the cellular SIM ICCID to the console
#       cell-signal
#         - prints the cellular signal level to the console
#
#######################################################################
```

Notes: